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Vector Spaces and Subspaces Graded Assignment

Graded Assignment • 10 min

Due Nov 13, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. **True** or **False**: If *S* is a subspace of the polynomial space $\mathbf{P}^n$, then $S = \mathbf{P}^m$ for some *m* with $0 \leq m \leq n$. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work

2. **True** or **False**: If *V* is a vector space and $\mathbf{v} + \mathbf{w} = \mathbf{0}$ for all vectors **v** and **w** in *V*, then *V* consists of only the zero vector. 1 / 1 point

   - ◉ True
   - ○ False

   Nice work

3. **True** or **False**: For every $n \in \mathbb{N}$, the set $\mathbf{P}^n$ of all polynomials of degree less than or equal to *n* is a subspace of $C(\mathbb{R})$. 1 / 1 point

   - ◉ True
   - ○ False

   Nice work

4. **True** or **False**: If $m \leq n$ then the polynomial space $\mathbf{P}^m$ is a subspace of $\mathbf{P}^n$. 1 / 1 point

   - ◉ True
   - ○ False

   Nice work

5. **True** or **False**: The set *V* of all nonnegative real numbers, using the usual rules for vector addition and scalar multiplication in R, is a vector space. 1 / 1 point

   - ○ True
   - ◉ False

   Nice work